

Asymptotically good construction of EAQECC with controlled amount of entanglement

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Abstract

We show a construction of asymptotically good linear codes with controlled dimensions of Euclidean hulls. By using it and Steane's enlargement construction for EAQECCs (entanglement-assisted quantum error-correcting codes), we propose a construction of asymptotically good EAQECCs with a controllable amount of entanglement, in contrast to the previous asymptotic constructions of EAQECCs whose amounts of entanglement were fixed to the maximal, while a smaller amount of entanglement is generally preferred in quantum error correction.

Key Words: Entanglement-assisted quantum error correction, asymptotic construction, algebraic geometry code

1 Introduction

Explicit constructions of asymptotically good linear codes have been an important topic in coding theory. In 1982, Tsfasman, Vlăduț and Zink showed the existence of asymptotically good algebraic geometry codes exceeding the Gilbert-Varshamov bound over large finite fields. Their bound on asymptotically good linear codes is now called the TVZ bound. For quantum error correction, also by using algebraic geometry codes, several asymptotically good constructions were proposed [3, 10, 23]. The asymptotic constructions of quantum codes were improved by concatenation [13, 18] over certain ranges of parameters.

It is now well-known that quantum error-correcting codes can be constructed from self-orthogonal (or dual-containing) linear spaces over finite fields [3, 5, 6,

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16, 19]. In general, the self-orthogonal constraint is difficult to be satisfied. To overcome this difficulty, the entanglement-assisted quantum error-correcting codes (EAQECCs) were proposed [4] for qubits and later extended to qudits [22]. An EAQECC can be constructed from any linear code with no self-orthogonality constraint. In an EAQECC, its encoder and decoder share c pairs of a maximally entangled state. The relation between the amount c of entanglement and its underlying linear code had been somewhat unclear, but in [14, 17], the relation $c = \dim C - \dim C \cap C^\perp$ was established, where C is a classical linear code defining an EAQECC and $\perp$ denotes taking the dual code of C with respect to an inner product. The hull (with respect to a chosen inner product) of C is $C \cap C^\perp$, and its dimension $\dim C \cap C^\perp$ is called the hull dimension. When the hull dimension is zero, the corresponding EAQECC is said to have maximal entanglement (ME), while the maximal hull dimension $\dim C = \dim C \cap C^\perp$ corresponds to a quantum error-correcting code with no entanglement. It was also revealed that the hull dimension can be decreased [1, 9, 21] and increased under certain conditions [1] by replacing the original codes with their equivalent codes while preserving code parameters. Starting from an asymptotically good self-orthogonal code $C \subseteq C^\perp$ [28], one can find an equivalent code $(C')^\perp$ with the same parameters as $C^\perp$ and an arbitrarily chosen amount of entanglement $\dim C' - \dim C' \cap (C')^\perp$. As we will show in Section 4, the parameters of an EAQECC given by C' are strictly worse than our construction. The superiority of our construction comes from the use of Steane's enlargement construction for EAQECCs [15], which requires some conditions on linear codes that are not ensured by merely finding an equivalent code C' as above. In this sense our proposed construction is not a trivial application of [28] or [1, 9, 21]. For the same reason, our proposed construction cannot be deduced as a straightforward combination of [24] and [1].

Recently, the first asymptotically good construction of EAQECCs was proposed [24], and later improved by concatenation [12]. Both constructions [12, 24] had maximal entanglement and could not adjust their amounts of entanglement to values other than the maximal. Since the generation and preservation of shared entanglement are difficult within the current technology [30], shared entanglement between an encoder and a decoder is a costly and precious resource in EAQECCs. Therefore, it is desirable to have an asymptotically good construction with a controllable amount of entanglement.

It seems difficult to construct asymptotically good linear codes attaining the TVZ bound with control of their hull dimensions while also allowing for self-orthogonal or dual-containing codes. In [23], TVZ-attaining asymptotically good self-orthogonal linear codes were explicitly shown and they were used for construction of asymptotically good binary quantum codes. The construction [23] did not work when the characteristic of finite fields is odd. Later Stichtenoth showed TVZ-attaining asymptotically good self-orthogonal linear codes over every finite field with square number of elements [28]. We will use algebraic function fields and their differentials in [28], and modify construc-

tions of linear codes in [28]. Our modified linear codes will attain the TVZ bound asymptotically and will also enable complete control over their hull dimensions with respect to the Euclidean inner product, as shown in this extended abstract.

In the previous constructions of asymptotically good quantum codes [3, 10, 23], Steane's enlargement construction [11, 20, 26] tended to provide better asymptotic parameters of quantum codes than the CSS construction [7, 25]. We will propose asymptotically good EAQECCs with control over the amount of entanglement by the TVZ-attaining asymptotically good linear codes and Steane's enlargement construction for EAQECCs [15], and evaluate their asymptotic parameters. This extended abstract is organized as follows: In Section 2, we review previous research results that will be used later. In Section 3, we construct asymptotically good linear codes with control over their Euclidean hull dimension. After that, by using them and Steane's enlargement construction, we propose an asymptotically good construction of EAQECCs with controlled amount of entanglement. In Section 4, we clarify asymptotic parameters of EAQECCs in Section 3.

2 Preliminaries

2.1 Review of Steane's enlargement for EAQECCs

In this extended abstract, the inner product is the Euclidean unless otherwise specified, and $C^\perp$ denotes the dual code of a linear code C with respect to the Euclidean inner product, while the symplectic inner product will be used in Section 2.2 below. Let q be a prime power, and by $\mathbb{F}_q$ we denote the finite field with q elements.

The following is a review of [15, Section 2.1]. Let C be an $[n, k]_q$ linear code and $c = \dim C - \dim(C \cap C^\perp)$. One can construct an

$$[[n, n + c - 2k, w_H(C^\perp \setminus C); c]]_q \quad (1)$$

EAQECC from C , where w_H denotes the minimum Hamming weight of nonzero vectors in a set.

Choose $C_t \subseteq C \cap C^\perp$ such that $\dim C_t \geq 2$ and decompose $C = C_r \oplus C_t$ as a direct sum of linear spaces. Let B_r be a generator matrix of C_r and B_t be a generator matrix of C_t in [15, Theorem 7]. Then by [15, Theorem 7], we can have an EAQECC with parameter

$$[[n, n + c - 2 \dim C_r - \dim C_t, d'; c]]_q, \quad (2)$$

where

$$d' \geq \min \left\{ w_H(C^\perp), \left\lceil \frac{1+q}{q} w_H(C_r^\perp) \right\rceil \right\}. \quad (3)$$

Observe that the number of information symbols is increased by $\dim C_t$.

The number of information symbols $n + c - 2 \dim C_r - \dim C_t$ is maximized by taking $C_t = C \cap C^\perp$. In such a case we have $\dim C_r = c$, $\dim C_t = k - c$, and (2) becomes

$$[[n, n - k, d'; c]]_q. \quad (4)$$

2.2 Converting codes over $\mathbf{F}_{q^m}$ to those over $\mathbf{F}_q$

In the constructions of asymptotically good binary quantum codes without entanglement [3, 10, 23], linear codes over $\mathbf{F}_{q^m}$ are converted to those over $\mathbf{F}_q$ while preserving the Euclidean inner product. We will also use that technique and review it in this section. In the construction of the EAQECC in (1) from the linear code $C \subset \mathbf{F}_q^n$, one considers $C \times C \subset \mathbf{F}_q^{2n}$, then uses $C \times C$ for defining a group of Pauli matrices on $\mathbf{C}^{q^n}$.

For vectors $\mathbf{x} = (x_1, \dots, x_{2n})$ and $\mathbf{y} = (y_1, \dots, y_{2n}) \in \mathbf{F}_q^{2n}$, in Section 2.2 we consider their symplectic inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle_s = \sum_{i=1}^n (x_i y_{n+i} - x_{n+i} y_i), \quad (5)$$

and the symplectic weight

$$w_s(\mathbf{x}) = |\{i \in \{1, \dots, n\} : (x_i, x_{n+i}) \neq (0, 0)\}|.$$

For an $\mathbf{F}_q$ -linear code $C_s \subset \mathbf{F}_q^{2n}$, by $C_s^{\perp_s}$ we denote the orthogonal space of the code C_s in $\mathbf{F}_q^{2n}$ with respect to the symplectic inner product (5). Then C_s defines an $[[n, n - \dim C_s + \frac{1}{2}(\dim C_s - \dim C_s \cap C_s^{\perp_s}), w_s(C_s^{\perp_s} \setminus C_s); \frac{1}{2}(\dim C_s - \dim C_s \cap C_s^{\perp_s})]]_q$ EAQECC, where $w_s(C_s^{\perp_s} \setminus C_s)$ denotes the minimum symplectic weight $\min\{w_s(\mathbf{x}) : \mathbf{x} \in C_s^{\perp_s} \setminus C_s\}$. [14].

Let $C_{s,m} \subset \mathbf{F}_{q^m}^{2n}$ be an $\mathbf{F}_{q^m}$ -linear space. By using [2] or [14], from $C_{s,m}$ one can construct an $[[nm, m(n - \dim C_{s,m} + \frac{1}{2}(\dim C_{s,m} - \dim C_{s,m} \cap C_{s,m}^{\perp_s})), w_s(C_{s,m}^{\perp_s} \setminus C_{s,m}); \frac{m}{2}(\dim C_{s,m} - \dim C_{s,m} \cap C_{s,m}^{\perp_s})]]_q$ EAQECC, where the dimensions are counted as $\mathbf{F}_{q^m}$ -linear spaces. The result in Section 2.2 will be useful in Section 4 when it is combined as a kind of concatenation for the construction of q -ary quantum codes with a small or non-square q .

3 Asymptotically good linear codes with control over their hull dimensions

3.1 Towers of algebraic function fields suitable for our construction

We need a tower of algebraic function fields on which algebraic geometry codes allow us to control their hull dimension. One can use [23] for even characteris-

tic, but it does not work for odd characteristic. We will use [28] as it works for every positive characteristic.

We will review [28]. Hereafter q is assumed to be a square in addition to being a prime power. Let $F_0 = \mathbf{F}_q(x_0)$ be the rational function field. For $i \geq 0$, let $F_{i+1} = F_i(x_{i+1})$ with

$$x_{i+1}^{\sqrt{q}} + x_{i+1} = \frac{x_i^{\sqrt{q}}}{x_i^{\sqrt{q}-1} + 1}.$$

We define

$$w = x_0^{\sqrt{q}} + x_0, \quad z = w^{\sqrt{q}-1}.$$

We now define a tower $E_0 \subset E_1 \subset \dots$ of algebraic function fields. Let $E_0 = \mathbf{F}_q(z)$, and for $i \geq 1$, E_i is defined as the Galois closure of F_{i-1}/E_0 , that is, the Galois closure of $\mathbf{F}_q(x_0, x_1, \dots, x_{i-1})/\mathbf{F}_q(z)$.

Hereafter we fix a positive integer i , and divisors and valuations are always considered in $E_i/\mathbf{F}_q$ unless otherwise stated. We will use the following properties of $E_i/\mathbf{F}_q$.

Proposition 1 [28] *For $\alpha \in \mathbf{F}_q$, by Q_α we denote the zero of $z - \alpha$ in $E_0 = \mathbf{F}_q(z)$, and by Q_∞ the pole of z in E_0 . Define a differential η by*

$$\eta = \frac{dw}{1-z}.$$

Finally by n_i we denote the extension degree of E_i/E_0 . We have the following properties:

1. *The extension degree n_i of E_i/E_0 is $(\sqrt{q}-1)\sqrt{q}^i p^{t(i)}$, where $t(i)$ is a nonnegative integer and p is the characteristic of $\mathbf{F}_q$.*
2. *The place Q_1 of E_0 splits completely in E_i/E_0 . We denote places of E_i lying over Q_1 as $P_1, \dots, P_{n_i}$.*
3. *There is a unique place P_∞ of degree one in E_i lying over Q_∞ , and also there is at least one place P_0 of degree one in E_i lying over Q_0 .*
4. *Let g_i be the genus of E_i . Then n_i/g_i converges to $\sqrt{q}-1$ as $i \rightarrow \infty$.*
5. *There exists a divisor $G_0 \geq 0$ of E_i such that $(\eta) = 2G_0 - (P_1 + \dots + P_{n_i})$, which implies $\deg G_0 = g_i - 1 + n_i/2$.*
6. *For $1 \leq j \leq n_i$, we have $\text{res}_{P_j}(\eta) \in \mathbf{F}_{\sqrt{q}} \setminus \{0\}$.*

For a divisor G of E_i whose support does not contain any of the P_j , we define

$$C(G) = \{(f(P_1), \dots, f(P_{n_i})) : f \in \mathcal{L}(G)\}.$$

Let $\mathbf{b} = (b_1, \dots, b_{n_i}) = (\text{res}_{P_1}(\eta), \dots, \text{res}_{P_{n_i}}(\eta)) \in (\mathbf{F}_{\sqrt{q}} \setminus \{0\})^{n_i}$, and fix $\mathbf{a} = (a_1, \dots, a_{n_i}) \in \mathbf{F}_q^{n_i}$ such that the component-wise product (Schur product) of $\mathbf{a}$ with itself is $\mathbf{b}$, that is,

$$a_i^2 = b_i \quad (6)$$

for $i = 1, \dots, n_i$. Note that there are 2^{n_i} choices of $\mathbf{a}$ for odd q , while $\mathbf{a}$ is unique for even q .

The following proposition was essentially given in [28].

Proposition 2 *For any divisor G with $\dim \mathcal{L}(G) \geq 1$, whose support does not contain any of the P_j , we have $(\mathbf{a} \cdot C(G_0 + G))^\perp = \mathbf{a} \cdot C(G_0 - G)$, where for a set $C \subset \mathbf{F}_q^{n_i}$, $\mathbf{a} \cdot C$ denotes the set of component-wise products of $\mathbf{a}$ and $\mathbf{c} \in C$.*

Proof. This is essentially shown in [28], but here a proof is given for completeness. By [27, Corollary 2.7] we have

$$\begin{aligned} C(G_0 + G)^\perp &= \mathbf{b} \cdot C(P_1 + \dots + P_{n_i} + (\eta) - G - G_0) \\ &= \mathbf{b} \cdot C(G_0 - G). \end{aligned}$$

The relation (6) between $\mathbf{a}$ and $\mathbf{b}$ implies $C(G_0 + G)^\perp = \mathbf{a} \cdot \mathbf{a} \cdot C(G_0 - G)$, which completes the proof. ■

3.2 Asymptotically good EAQECs with controlled amount of entanglement

Let ℓ_0, ℓ_∞ be nonnegative integers. We will use $\mathbf{a} \cdot C(G_0 - \ell_\infty P_\infty + \ell_0 P_0)$ as C in Section 2.1. Then by Proposition 2, we have

$$\begin{aligned} C^\perp &= \mathbf{a} \cdot C(G_0 + \ell_\infty P_\infty - \ell_0 P_0), \\ C \cap C^\perp &= \mathbf{a} \cdot C(G_0 - \ell_\infty P_\infty - \ell_0 P_0). \end{aligned}$$

Let

$$\ell_r \geq 2 + \ell_\infty. \quad (7)$$

If we use $\mathbf{a} \cdot C(G_0 - \ell_r P_\infty + \ell_0 P_0)$ as C_r in Section 2.1, then we have $C_r^\perp = C(G_0 + \ell_r P_\infty - \ell_0 P_0)$, and we use an $(\ell_r - \ell_\infty)$ -dimensional space contained in $\{\mathbf{0}\} \cup (\mathbf{a} \cdot C(G_0 - \ell_\infty P_\infty - \ell_0 P_0) \setminus \mathbf{a} \cdot C(G_0 - \ell_r P_\infty - \ell_0 P_0))$ as C_t .

Proposition 3 *Let $f_j \in \mathcal{L}(G_0 - (\ell_r - j)P_\infty - \ell_0 P_0) \setminus \mathcal{L}(G_0 - (\ell_r - j + 1)P_\infty - \ell_0 P_0)$ for $j = 1, \dots, \ell_r - \ell_\infty$, and let C'_t be a linear space spanned by $(f_j(P_1), \dots, f_j(P_{n_i}))$ for $j = 1, \dots, \ell_r - \ell_\infty$, and $C_t = \mathbf{a} \cdot C'_t$. If*

$$2g_i - 1 \leq \deg(G_0 - \ell_r P_\infty - \ell_0 P_0), \quad (8)$$

$$n_i - 1 \geq \deg(G_0 - \ell_\infty P_\infty + \ell_0 P_0), \quad (9)$$

then we have

$$C_t \cap C_r = \{\mathbf{0}\}, \quad (10)$$

$$C_t + C_r = C. \quad (11)$$

Proof. By (9), evaluation of functions in $\mathcal{L}(G_0 - \ell_\infty P_\infty + \ell_0 P_0)$ is a linear isomorphism to C . By (8), for all $1 \leq j \leq \ell_r - \ell_\infty$, we have $\mathcal{L}(G_0 - (\ell_r - j)P_\infty - \ell_0 P_0) \neq \mathcal{L}(G_0 - (\ell_r - j + 1)P_\infty - \ell_0 P_0)$. So we can define $f_j \in \mathcal{L}(G_0 - (\ell_r - j)P_\infty - \ell_0 P_0) \setminus \mathcal{L}(G_0 - (\ell_r - j + 1)P_\infty - \ell_0 P_0)$ as above for $j = 1, \dots, \ell_r - \ell_\infty$, and we also have $v_{P_\infty}(f_j) = \ell_r - j - v_{P_\infty}(G_0)$. By the strict triangle inequality of the valuation at P_∞ [29], $f_1, \dots, f_{\ell_r - \ell_\infty}$ are linearly independent modulo $\mathcal{L}(G_0 - \ell_r P_\infty + \ell_0 P_0)$. Then $C_t \cap C_r = \{\mathbf{0}\}$. By (9) we see $\dim C - \dim C_r = \ell_r - \ell_\infty$, which is equal to $\dim C_t$, which in turn implies $C = C_t + C_r$. ■

Theorem 4 Assume that

$$\deg(G_0 - \ell_\infty P_\infty + \ell_0 P_0) \leq n_i - 1, \quad (12)$$

$$\deg(G_0 - \ell_\infty P_\infty - \ell_0 P_0) \geq 2g_i - 1, \quad (13)$$

$$\deg(G_0 - \ell_r P_\infty + \ell_0 P_0) \geq 2g_i - 1. \quad (14)$$

By using C, C_r, C_t defined above, and the method in Section 2.1, one can construct an EAQECC with parameters $[[n_i, k_i, d_i; c_i]]_q$ with

$$\begin{aligned} c_i &= \dim C(G_0 - \ell_\infty P_\infty + \ell_0 P_0) - \dim C(G_0 - \ell_\infty P_\infty - \ell_0 P_0) \\ &= 2\ell_0, \\ k_i &= c_i + n_i - 2 \dim C_r - \dim C_t \\ &= c_i + n_i - 2(g_i - 1 + n_i/2 - \ell_r + \ell_0 - g_i + 1) - (\ell_r - \ell_\infty) \\ &= \ell_r + \ell_\infty, \\ d_i &\geq \min \left\{ w_H(C^\perp), \left\lceil \frac{q+1}{q} w_H(C_r^\perp) \right\rceil \right\} \\ &= \min \left\{ w_H(C(G_0 + \ell_\infty P_\infty - \ell_0 P_0)), \left\lceil \frac{q+1}{q} w_H(C(G_0 + \ell_r P_\infty - \ell_0 P_0)) \right\rceil \right\} \\ &\geq \min \left\{ n_i - (g_i - 1 + n_i/2 + \ell_\infty - \ell_0), \right. \\ &\quad \left. \left\lceil \frac{q+1}{q} [n_i - (g_i - 1 + n_i/2 + \ell_r - \ell_0)] \right\rceil \right\} \\ &= \min \left\{ 1 + n_i/2 - g_i - \ell_\infty + \ell_0, \left\lceil \frac{q+1}{q} (1 + n_i/2 - g_i - \ell_r + \ell_0) \right\rceil \right\}. \end{aligned} \quad (15)$$

By considering $\deg G_0 = n_i/2 + g_i - 1$ and (7)–(9), the condition

$$\ell_r + \ell_0 \leq \frac{n_i}{2} - g_i \quad (17)$$

is sufficient for (12)–(14) to hold. ■

4 Asymptotic parameters of the proposed construction

In [3, 10, 23], asymptotically good binary quantum codes were constructed. Since 2 is not a square prime power, Theorem 4 or its no-entanglement version cannot

be used directly to construct binary quantum codes. In [3, 23], firstly one constructs asymptotically good self-orthogonal codes over some larger field, then those self-orthogonal codes are converted to binary self-orthogonal codes by the result in Section 2.2, and finally Steane's enlargement [11, 20, 26] is used to obtain binary quantum codes. We will follow a similar path.

Suppose that we will construct a q_0 -ary asymptotically good EAQECC, for a prime power q_0 . For some positive integer m , let $q = q_0^m$ be a square. Then, construct q -ary linear codes C , C_r , C_t as in the last section, construct an $\mathbf{F}_q$ -linear space D by using [15] with C , C_r , C_t , and finally apply the result in Section 2.2 to D which is used as $C_{s,m}$ in Section 2.2. Quantum codes of the following parameters can be obtained.

Theorem 5 *If $\ell_r + \ell_0 \leq \frac{n_i}{2} - g_i$, by using C , C_r , $C_t \subset \mathbf{F}_q^{n_i}$ in Section 3 and the result in Sections 2.1 and 2.2, one can construct an EAQECC with parameters $[[mn_i, mk_i, d_i; mc_i]]_{q_0}$ with*

$$\begin{aligned} c_i &= 2\ell_0, \\ k_i &= \ell_r + \ell_\infty, \\ d_i &\geq \min \left\{ 1 + n_i/2 - g_i - \ell_\infty + \ell_0, \right. \\ &\quad \left. \left\lceil \frac{q_0 + 1}{q_0} (1 + n_i/2 - g_i - \ell_r + \ell_0) \right\rceil \right\}. \end{aligned} \quad (18)$$

■

Let δ'/m be a desired relative quantum distance, $\gamma = g_i/n_i$ be the genus relative to the number of evaluation points, and θ be the ratio of entanglement divided by the code length mn_i . For the construction of asymptotically good EAQECCs, parameters are chosen as follows, ignoring fractional parts of real numbers.

$$\ell_0 = n_i \theta / 2, \quad (19)$$

$$\ell_\infty = n_i \left(\frac{1}{2} + \frac{\theta}{2} - \gamma - \delta' \right), \quad (20)$$

$$\ell_r = n_i \left(\frac{1}{2} + \frac{\theta}{2} - \gamma - \frac{q_0 \delta'}{q_0 + 1} \right). \quad (21)$$

By the above choices, the information rate k_i/n_i is

$$\frac{k_i}{n_i} = 1 + \theta - 2\gamma - \frac{2q_0 + 1}{q_0 + 1} \delta'.$$

Consider asymptotically good algebraic geometry codes C_i with parameters $[n_i, n_i - g_i - n_i \delta', n_i \delta']_q$, which attain the TVZ bound, and hull dimension $n_i \theta = \dim C_i^\perp - \dim C_i^\perp \cap C_i$. The CSS construction with C_i and the result in Section 2.2 (originally given in [2]) give an EAQECC with information rate $1 + \theta - 2\gamma - 2\delta'$, relative

quantum distance δ'/m and relative consumption of entanglement θ . We observe that the information rate of the proposed construction is larger by $\frac{1}{q_0+1}\delta'$ than the straightforward application of the combination of [28] and [1,9,21], or the combination of [24] and [1]. The increase $\frac{1}{q_0+1}\delta'$ comes from the use of [15].

We will clarify the possible ranges of θ and relative distance $\delta = \delta'/m$. By (17), (19) and (21), the possible range of θ is $[0, q_0\delta'/(q_0 + 1)]$. By (20) and $\ell_\infty \geq 0$, we have $\delta' \leq \frac{1+\theta}{2} - \gamma$. The possible ranges of θ and δ are

$$0 \leq \frac{q_0 + 1}{mq_0}\theta \leq \delta \leq \frac{1 + \theta}{2m} - \frac{\gamma}{m}, \quad (22)$$

which is consistent with [23] with $\theta = 0$. Note that in [23] we have $q_0 = 2$ and $q = 2^{2m}$.

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